

TECHNICAL LEAD - SOLUTION ARCHITECT - SENIOR DEVELOPER

Juan van der Merwe

Software, AI & Integration

Enterprise software professional with more than 13 years of experience across MES, solution architecture, real-time data platforms, enterprise integration, AI initiatives and technical leadership.

13+

Years' experience

4-6

Developer team

MES

Mining & operations

AI

Agents, RAG & local LLMs

PROFESSIONAL PROFILE

Hands-on leadership grounded in architecture and delivery

Senior software professional with extensive experience in enterprise software development, Manufacturing Execution Systems (MES), solution architecture, systems integration and real-time operational data platforms within mining and industrial environments.

Connects historian and enterprise data to operational intelligence, time-series reporting and decision-making.

Hands-on focus on enterprise integrations, intelligent reporting and AI-enabled solutions.

Leadership focus

- Technical strategy and governance
- Architecture standards and code reviews
- Developer mentorship and support
- AI adoption and responsible governance
- Client engagement and roadmap planning

Core competencies

- Solution architecture and integration
- Manufacturing Execution Systems
- Real-time data and time-series architectures
- Event-driven systems and APIs

CAREER PROGRESSION

Career progression

MINOPEX | JUN 2021 - PRESENT

Senior Developer

Technical lead for Minocore (MES) and MTAC (Operational Readiness), with responsibility across architecture, integrations, platform modernisation, developer support and strategic initiatives.

Leadership & governance

- Lead technical delivery across a team of 4-6 developers.
- Provide architecture guidance, code reviews, mentoring and governance.
- Support third-line tasks, solution configuration and technical delivery.
- Support the Head of IT with strategy and roadmap decisions.

Architecture & modernisation

- Lead the technical evolution of Minocore and MTAC.
- Lead the Vue 2 to Vue 3 modernisation programme.
- Architect Sage, Vendor Portal and Neuromine solutions.
- Define development, architecture and integration standards.

Real-time operational data architecture

Industrial Historians -> Real-time Ingestion -> TimescaleDB -> Aggregation Services -> Minocore MES

TimescaleDB powers Grafana trending, operational analysis and monitoring dashboards.

AI initiatives

MTAC Gap Assessment AI Agent, OpenAI integration, RAG, embeddings, reranking and local AI infrastructure using the NVIDIA GB10 platform.

EARLIER CAREER

SIBANYE-STILLWATER / LONMIN PGM | JUN 2018 - MAY 2021

Application Analyst Developer

- Material Tracking, Metals Accounting, Asset Monitoring and OEE solutions.
- MES integrations with Wonderware Historian, SAP ERP and LIMS.
- Mentoring junior developers and data-science resources.

EOH MICROSOFT SERVICES | FEB 2012 - MAY 2018

Business Analyst -> Systems Analyst -> Senior Developer

Consulting delivery for Liberty, Famous Brands, FNB Home Loans, FNB Private Clients, RMB and Nedbank, spanning business analysis, solution design, enterprise integration, document management and Sage ERP integrations.

PROJECTS, TECHNOLOGY & ACADEMIC FOUNDATION

SELECTED PROJECTS

Architecture, integration and innovation

OPERATIONAL READINESS

MTAC Platform & AI Agent

Platform architecture, delivery, solution ownership, AI-assisted gap assessment and modernisation.

MANUFACTURING EXECUTION

Minocore MES

MES architecture, OEE, historian data ingress, operational intelligence and Vue modernisation.

ENTERPRISE INTEGRATION

Sage, OnKey & Neuromine

Business-critical integration patterns, SSO and information services.

INDEPENDENT PROJECT

Energy Management Platform

ASP.NET web application, REST APIs, .NET MAUI, Firebase, MQTT services and telemetry.

INDEPENDENT PROJECT

Expo React Native PoC

Completed and approved for further development with notifications, offline sync and API integration.

TECHNOLOGY STACK

Broad technical depth

Languages

C#, SQL, TypeScript, JavaScript, Python

Frameworks

.NET, ASP.NET, Vue 2, Vue 3, MAUI, React Native, Expo

Data platforms

SQL Server, PostgreSQL, TimescaleDB, historian systems

Industrial systems

MES, SAP ERP, Wonderware Historian, LIMS

Architecture

REST APIs, MQTT, event-driven, SSO, SAML, offline sync

Analytics

Grafana, Power BI, SSRS, operational intelligence

AI

OpenAI, RAG, embeddings, reranking, local LLMs, governance

Delivery

Docker, IIS, Firebase, Expo Notifications, NVIDIA GB10

EDUCATION & CERTIFICATION

Academic foundation

BCom (Honours) Informatics

University of Pretoria - 2013

BCom Informatics

University of Pretoria - 2011

Certifications

ITIL v3 | Microsoft HTML5, JavaScript & CSS3 | WPF Fundamentals

CONTINUOUS LEARNING & INNOVATION

Professional development

Completed

Vue - The Complete Guide

In progress

Vue - Supabase | AI Engineer

Areas of focus

AI, modern front-end architecture, cloud-native development, RAG and developer productivity